



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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LectureNotes-2.3

22GE004–BASICS OF ELECTRONICS ENGINEERING

UNIT II & SIGNAL CONDITIONING USING DIODE

Review of Semiconductor Physics

1. Energy Bands in Materials

1.1. Valence Band

- The outermost electrons of an atom i.e. those in the shell furthestmost from the nucleus are called valence electrons and have the highest energy
- However, their binding energy to the nucleus is the least
- These electrons are affected when the atom comes close to other atoms
- The band of energy occupied by these valence electrons is called the valence band
- It may be completely filled or partially filled with electrons but never empty
- Valence band is the highest occupied band

1.2. Conduction band

- The next higher band to Valence band is called Conduction band
- It may be empty or partially filled with electrons
- In conduction band, electrons can move freely and hence are known as conduction band
- The gap between Valence band and Conduction band is known as the forbidden energy gap.

1.3. The electrical conduction mechanism in materials

- If a valence electron happens to absorb enough energy, it jumps across the forbidden energy gap and enters the conduction band
- If the electron gains more energy, it can jump to the next higher conduction bands
- If the electron loses energy, it may come back to the valence band
- When an electron is ejected from the valence band, a covalent bond is broken and a positively charged hole is left behind
- This mechanism is shown in Figure 1

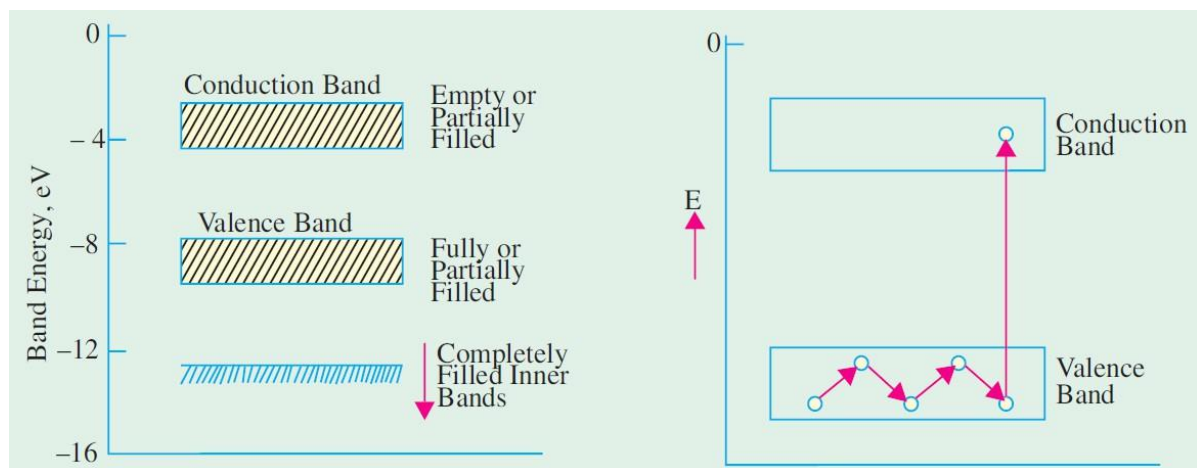


Figure 1. Energy Bands in Materials

- This hole can travel to an adjacent atom by acquiring an electron from that atom which involves breaking an existing covalent bond and then re-establishing a covalent bond by filling up the hole
- The holes are filled by electrons which move from adjacent atoms without passing through the forbidden energy gap.
- The free electrons are flowing in the conduction band. However, conduction band have nothing to do with the hole flow
- Although holes flow with ease, they experience more opposition than electron flow in the conduction band
- In materials, Conduction electrons is twice faster compared to the conduction of holes

2. Classification of Materials based on Electrical Conductivity

- The electrical conduction properties of different elements and compounds can be explained in terms of the electrons having energies in the valence and conduction bands.
- The electrons lying in the lower energy bands, which are normally filled, play no part in the conduction process.
- However, the Valance band, Conduction band and the Forbidden Energy Gap in the materials play a major role in the Electrical Conduction
- Based on Electrical Conduction properties, materials can be classified into three;
- They are
 - (a) Insulators
 - (b) Conductors
 - (c) Semiconductors

2.1. Insulators(Eg.Glass,Mica,Paper)

- insulators are those materials in which valence electrons are bound very tightly to their parents atoms, thus requiring very large electric field to remove them from the attraction of their nuclei.
- insulatorshavenofreechargecarriersavailablewiththemundernormalconditions
- IntermsofEnergyBands,theimportantpointstobenotedaboutInsulatorsare;
 - TheyhaveafullyfilledValanceBand
 - TheyhaveanemptyConductionBand
 - Theyhavealargeenergygap(ofseveraleV)betweenthem
 - At ordinary temperature, electrons in the valance bands gaining enough energy and jumping to the conduction band is almost impossible
- Thisisdiagramaticallyshowninthebelowfigure

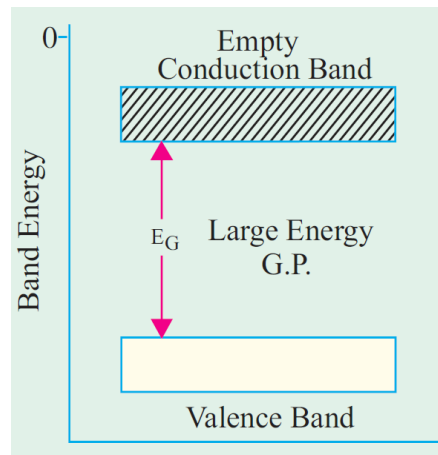


Figure2. Energy Bands in Insulators

2.2. Conductors (Eg. Copper, Aluminium, Silver)

- Conducting materials are those in which plenty of free electrons are available for electric conduction
- In terms of energy bands, it means that electrical conductors are those which have overlapping valence and conduction bands
- This is shown in the below Figure 2.

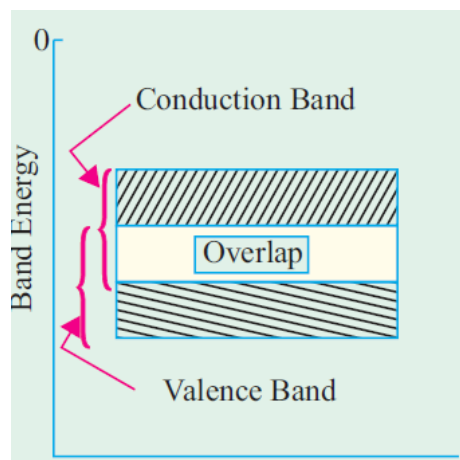


Figure3. Energy Bands in Conductors

- There is no forbidden energy gap present in conductors
- There is no structure present in conductors and the total current in conductors is simply because of the flow of electrons

2.3. Semiconductors (Eg. Silicon, Germanium)

- A semiconductor material is one whose electrical properties lie in between those of insulators and good conductors
- In terms of energy bands, semiconductors can be defined as those materials which have almost an empty conduction band and almost filled valence band with a very

narrow energy gap (of the order of 1 eV) separating the two.

- This is diagrammatically represented in the below figure.

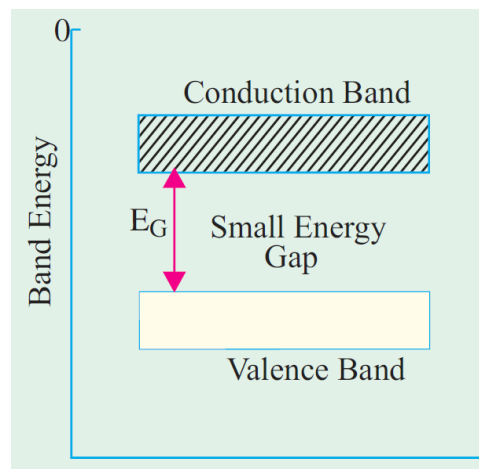


Figure 4. Energy Bands in Semiconductors

- At 0°K, there are no electrons in the conduction band and the valence band is completely filled.
- However, with increase in temperature, width of the forbidden energy bands is decreased so that some of the electrons are liberated into the conduction band.
- Conductivity of semiconductors increases with temperature.
- The departing electrons leave behind positive holes in the valence band.
- The mechanism of Electron Hole Pair generation in semiconductors under the influence of temperature is shown in the below figure.

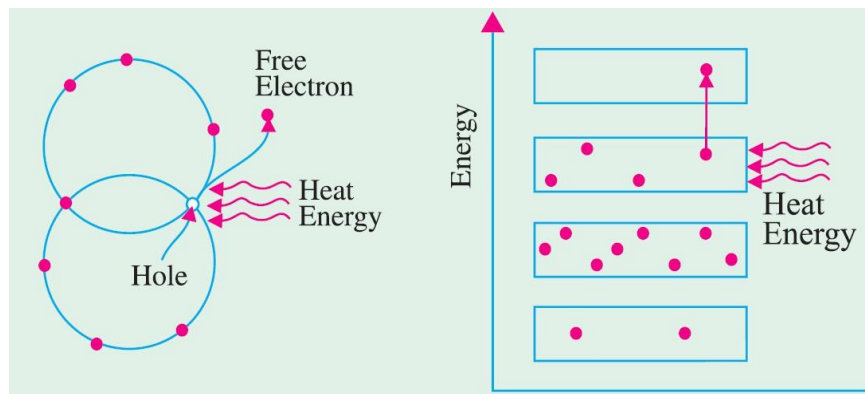


Figure 5. Electron Hole Pair Generation in Semiconductors

- Hence, semiconductor current is the sum of electron and hole currents flowing in opposite directions.

3. Classification of Semiconductors

- Based on Doping, the process of adding impurity elements into pure semiconductors to improve the properties, Semiconductors can be classified into

two types

- (a) Intrinsic Semiconductors (Pure Semiconductors)
- (b) Extrinsic Semiconductors (Doped Semiconductors)

3.1. Intrinsic Semiconductors

- An intrinsic semiconductor is one which is made of the semiconductor material in its extremely pure form
- In Intrinsic semiconductors, the energy gap is so small that even at ordinary room temperature, there are many electrons which possess sufficient energy to jump across the small energy gap between the valence and the conduction bands.
- For each electron liberated into the conduction band, a positively charged hole is created in the valence band
- Hence semiconductor current consists of movement of electrons and holes in opposite directions.
- Electron current is due to movement of electrons in the conduction band whereas hole current is within the valence band as a result of the holes 'jumping' from one atom to another.
- The number of charge carriers in Intrinsic Semiconductors per unit volume is given as

$$n_i = N e^{\left(\frac{-E_g}{2kT} \right)} \quad (1)$$

- Where,
 - N is a constant for a given semiconductor
 - E_g is the band gap energy in Joules
 - k is the Boltzmann's constant
 - T is the temperature in Kelvin

3.2. Extrinsic Semiconductors

- Those intrinsic semiconductors to which some suitable impurity or doping agent or dopant has been added in extremely small amounts (about 1 part in 10^8) are called extrinsic semiconductors
- The usual doping agents are;
 - (a) Pentavalent atoms having five valence electrons (arsenic, antimony, phosphorus)
 - (b) Trivalent atoms having three valence electrons (gallium, indium, aluminium, boron)
- Pentavalent doping atom is known as donor atom because it donates or contributes one electron to the conduction band of pure germanium.
- The trivalent atom, on the other hand, is called acceptor atom because it accepts one electron from the germanium atom.
- Depending on the type of doping material used, extrinsic semiconductors can be sub-divided into two classes
 - (c) P Type Semiconductors

(d) NTypeSemiconductors

3.2.1. PTypeSemiconductors

- This type of semiconductor is obtained when traces of a trivalent like Boron (B) are added to a pure germanium or Silicon crystal
- The three valence electrons of boron atom form covalent bonds with four surrounding germanium atoms but one bond is left incomplete and gives rise to a hole
- Since the Boron is making so many holes (which can accept electrons) in the Pure Semiconductor, they are called Acceptors
- In this type of semiconductor, conduction is by the movement of holes in the valence band.
- In P Type Semiconductors, holes form the majority carriers whereas electrons constitute minority carriers.
- The process of conduction is called 'deficit' conduction.
- Even though P-type semiconductor has excess of holes for conduction purposes, on the whole it is electrically neutral
- The process of Trivalent Doping is shown in the below figure.

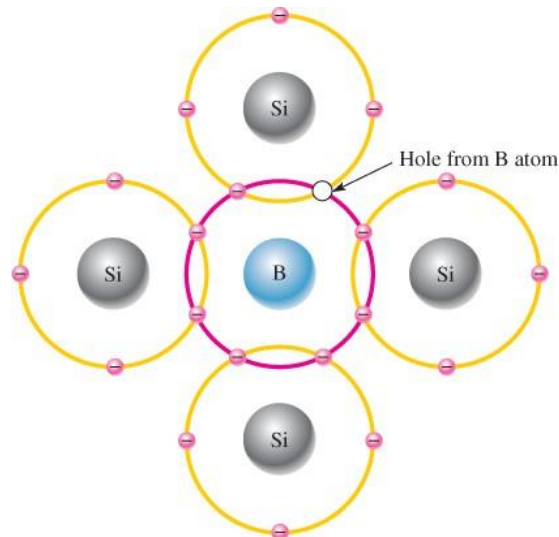


Figure6.TrivalentDopingofSiliconusingBoronAtom

3.2.2. NTypeSemiconductors

- This type of semiconductor is obtained when a pentavalent material like Antimony (Sb) is added to pure Germanium or Silicon crystal
- Each antimony atom forms covalent bonds with the surrounding four germanium atoms with the help of four of its five electrons.
- The fifth electron is superfluous and is loosely bound to the antimony atom.
- Hence, it can be easily excited from the valence band to the conduction band by the application of electric field or increase in thermal energy.
- Thus, practically every antimony atom introduced contributes one conduction electron without creating a positive hole.
- Antimony is called donor impurity and makes the pure germanium an N-type (N for

negative)extrinsicsemi-conductor.

- It is seen from the above description that in N-type semiconductors, electrons are the majority carriers while holes constitute the minority carriers.
- Hence, N-type semiconductor conducts principally by electrons in the nearly empty conduction band and the process is called 'excess' conduction.
- Even though N-type semiconductor has excess of electrons, still it is electrically neutral.
- TheprocessofPentavalentDopingisshowninthebelowFigure

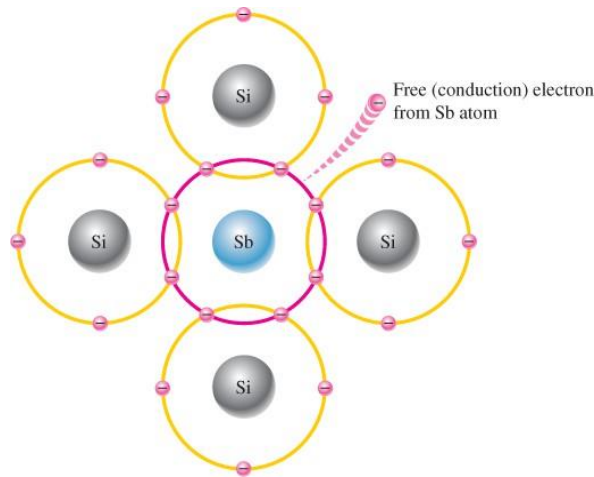


Figure.7PentavalentDopingofSiliconusingAntimonyatom

4. Charge Carrier Densities in Semiconductors

- The below table lists some typical quantities for intrinsic silicon and Germanium at room temperature of 300°K

Property	Silicon	Germanium
AtomicDensity(percm ³)	5×10^{22}	4.4×10^{22}
ElectronDensity n_i (percm) ³	1.5×10^{10}	2.5×10^{13}
HoleDensity p_i (percm) ³	1.5×10^{10}	2.5×10^{13}
ElectronMobility μ_n (cm ² /V.s)	1500	3800
HoleMobility μ_p (cm ² /V.s)	500	1800

4.1. MassActionLaw

- ConsidertheelectrondensityofSilicon,

$$n_i = 1.5 \times 10^{10} \quad (2)$$

- Thismeans that 1.5×10^{10} atoms haveeach released one elecron due tothermal

energy at room temperature of 27°C in the Silicon Material

- Consequently, 1.5×10^{10} holes also have been created.
- So the hole density p_i is the same as electron density n_i in Intrinsic Semiconductors
- Thus for an intrinsic semiconductor

$$n_i = p_i \quad (3)$$

- If a P Type Extrinsic Semiconductor has hole density p , some free electrons are absorbed into these holes.
- This will reduce the Intrinsic Free electron density n_i in the material
- Similarly if an N Type Extrinsic Semiconductor has an electron density n , some electrons will drop into the holes.
- This will reduce the Intrinsic Hole density p_i in the material
- This is known as "**Mass Action Law**"
- Mass action Law says that the hole-electron density product for extrinsic material equals that for Intrinsic material.
- That is,

$$pn = p_i n_i \quad (4)$$

- For Intrinsic Material, from equation (3), $n_i = p_i$
- Hence the equation will be modified to

$$pn = n_i^2 \quad (5)$$

- When an intrinsic material is doped with N_D donor atoms per cubic centimeter, each donor atom releases an electron so that the atom becomes a positive ion.
- This adds to the positive charge contributed by the hole density p_i
- So the total Positive Charge density is given as

$$\text{Positive Charge Density} = N_D + p_i \quad (6)$$

- Since the semiconductor is electrically neutral,

$$\text{Negative Charge Density} = N_D + p_i \quad (7)$$

- When an intrinsic material is doped with N_A acceptor atoms per cubic centimeter, each acceptor atom accepts an electron so that the atom becomes a negative ion.
- This adds to the negative charge contributed by the hole density n_i
- So the total Negative Charge density is given as

$$\text{Negative Charge Density} = N_A + n_i \quad (8)$$

- Since the semiconductor is electrically neutral,

$$\text{Positive Charge Density} = N_A + n_i \quad (9)$$

- If both donor and acceptor doping are involved,

$$N_A + p_i = N_A + n_i \quad (10)$$

Factors affecting the conductivity of intrinsic and extrinsic semiconductors

1. Electron Conductivity of a Metal

- In metals, the valance electrons in an atom are not attached to individual atoms.
- They are free to move in all directions among the atoms.
- They are known as free electron cloud.
- In copper, for example there is one electron in the valance shell.
- When no external field is applied to the metal, free electrons move randomly in all directions
- This is shown in figure (1)

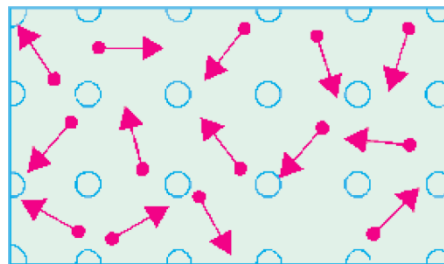


Figure 1. Random Movement of Electrons

- When an external electric field is applied to the metal, the free electron motion becomes directed as shown in figure (2)

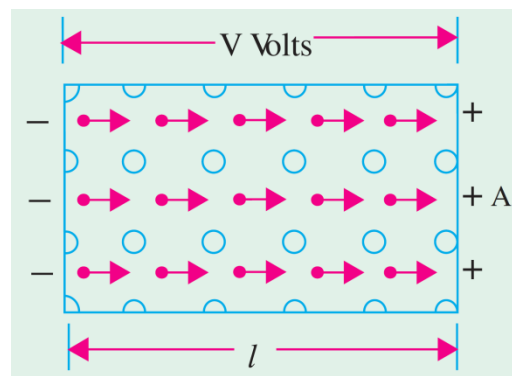


Figure 2. Drift Movement of Electrons in metals

- This directed flow of electrons results in a net charge displacement in a definite direction.
- This type of motion of electrons is known as drift.
- The drift velocity (v) of the electrons depends on the electron mobility μ_e and the applied electric field E .
- The drift velocity is given as

$$v = \mu_e E \quad (1)$$

- Where

$$\mu_e = \text{Mobility of Electrons}$$

$$E = \text{Applied Electric Field}$$

- Now we have to derive an expression for the current flowing through the metal.
- At any point the current is given by the total charge flowing through that point in one

second.

- The total charge carriers flowing through a point in one second depends on the total number of charge carriers n in unit volume (charge density) and the drift velocity v of the charge carriers.
- That is, for a point

$$\text{Total charge carriers/Second} = nv$$

- For a given cross-sectional area A ,

$$\text{Total charge carriers/Second} = nvA$$

- Now, for a given cross-sectional area,

$$\text{Total Charge/Second} = envA$$

- That is, the drift current is given as

$$I = envA \quad (2)$$

- Substituting (1) in (2)

$$I = en \square_e EA \quad (3)$$

- We know that the Electric Field is given by

$$E = \frac{V}{l} \quad (4)$$

- Substituting (4) in (3) will give

$$I = en \square_e \frac{V}{l} A \quad (5)$$

- The resistance of a metal is given as

$$R = \frac{V}{I} \quad (6)$$

- Substituting (5) in (6) will give

$$R = \frac{V}{en \square_e \frac{V}{l} A} = \frac{l}{A en \square_e} = \square \frac{l}{A} \quad (7)$$

- Thus the resistivity $\square$ is given as

$$\square = \frac{1}{en \square_e} \quad (8)$$

- The unit of $\square$ is ohm-m

- Conductivity of the metal $\square$, which is the reciprocal of Resistivity is given as

$$\square = en \square_e \quad (9)$$

- The unit of $\square$ is Siemens/m

- Using the equation (3), the current density in a metal can be written as

$$J = \frac{I}{A} = \frac{en \square_e EA}{A} = en \square_e E \quad (10)$$

- Using equation (9), we can rewrite the equation for current density as

$$J = \square E \quad (11)$$

2. Conductivity of Semiconductors

- In Semiconductors, the charge carriers constitute both electrons and holes, whereas in metals, electrons are the only charge carriers

2.1. Conductivity of Intrinsic Semiconductors

- In the case of Intrinsic Semiconductors, the conductivity is due to the movement of electrons and holes in opposite directions as shown in figure (3)

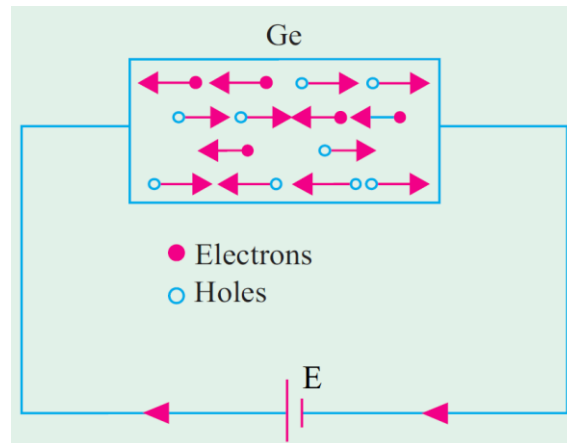


Figure3. Movement of Holes and Electrons in Semiconductors

- Since they have opposite charges, the current is in the same direction
- The number of holes and number of electrons are equal
- However, the mobility of holes μ_h is half that of the mobility of the electrons μ_e
- The total current is the sum of current due to holes and current due to electrons
- The total current in an intrinsic semiconductor is given as

$$I = I_h + I_e \quad (12)$$

- From Equation (2) we can write the current due to holes and electrons
- Current due to holes is given by

$$I_h = ep_i v_h A \quad (13)$$

- Here p_i is the density of holes in intrinsic semiconductors and v_h is the drift velocity of the holes
- Current due to electrons is given by

$$I_e = en_i v_e A \quad (14)$$

- Here n_i is the density of electrons in intrinsic semiconductors and v_e is the drift velocity of the electrons
- Substituting (14) and (13) in (12) we get

$$I = I_h + I_e = ep_i v_h A + en_i v_e A \quad (15)$$

- Since the hole concentration p_i is equal to the electron concentration n_i in intrinsic semiconductors, we can rewrite the equation (15) as

$$I = en_i A (v_h + v_e) \quad (16)$$

- From Equation (1) we can find the mobility of holes and electrons
- Thus the total current is given as

$$I = en_i A E (\mu_h + \mu_e) \quad (17)$$

- Substituting equation (4) in equation (17), we get

$$I = en_i A \frac{V}{l} (\mu_h + \mu_e) \quad (18)$$

- Thus the resistance of a semiconductor is given as

$$R = \frac{V}{I} = \frac{V}{en_i A \frac{V}{l} (\mu_h + \mu_e)} = \frac{l}{A en_i (\mu_h + \mu_e)} = \frac{l}{A \sigma_i} \quad (19)$$

- Thus the resistivity ρ_i is given as

$$\rho_i = \frac{l}{A \sigma_i} \quad (20)$$

- The electrical conductivity σ_i , which is the reciprocal of resistivity is given as

$$\sigma_i = en_i (\mu_h + \mu_e) \quad (21)$$

- From the above equation, it can be concluded that the conductivity of semiconductors depends on two factors
 - (a) Number of charge carriers present per unit volume
 - (b) Mobility of the charge carriers
- The increase in temperature will increase n_i and μ_i and thus the conductivity of intrinsic semiconductors increase with increase in temperature.
- From Equation (15), the current density can be written as

$$J = \frac{I}{A} = en_i E (\mu_h + \mu_e) \quad (22)$$

- Using equation (21), the equation for current density can be rewritten as

$$J = \sigma_i E \quad (23)$$

2.2. Conductivity of Extrinsic Semiconductors

- From equation (22) we can write a general expression for the current density in extrinsic semiconductors as

$$J = (p\mu_h + n\mu_e) e E \quad (24)$$

- Where p and n represent the carrier concentration of holes and electrons respectively in extrinsic semiconductors
- From (23) we can write a general expression for conductivity as

$$\sigma = \frac{J}{E} \quad (25)$$

- Substituting (24) in (25) we get

$$\sigma = \frac{J}{E} = \frac{(p\mu_h + n\mu_e) e E}{E} = (p\mu_h + n\mu_e) e \quad (26)$$

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2.2.1. Conductivity of P Type Semiconductors

- For a P type semiconductor, equation (24) will become

$$J = (p_p \mu_h + n_p \mu_e) e E \quad (27)$$

- Where p_p and n_p represent the carrier concentration of holes and electrons respectively in P Type semiconductor after doping
- From equation (26), the conductivity is given as

$$\sigma = e(p_p \mu_h + n_p \mu_e) \quad (28)$$

- In P type semiconductors, the holes are majority carriers and electrons are minority carriers.
- Hence we can assume that $p_p \gg n_p$
- In this case we can neglect the second term in equation (27) and (28)
- Equation (27) will be modified as

$$J = p_p \mu_h e E \quad (29)$$

- Equation (28) will be modified as

$$\sigma = e p_p \mu_h \quad (30)$$

2.2.2. Conductivity of N Type Semiconductors

- For an N type semiconductor, equation (24) will become

$$J = (p_n \mu_h + n_n \mu_e) e E \quad (31)$$

- Where p_n and n_n represent the carrier concentration of holes and electrons respectively in N Type semiconductor after doping
- From equation (26), the conductivity is given as

$$\sigma = e(p_n \mu_h + n_n \mu_e) \quad (32)$$

- In N type semiconductors, the electrons are majority carriers and holes are minority carriers.
- Hence we can assume that $n_n \gg p_n$
- In this case we can neglect the first term in equation (27) and (28)
- Equation (27) will be modified as

$$J = n_n \mu_e e E \quad (33)$$

- Equation (28) will be modified as

$$\sigma = e n_n \mu_e \quad (34)$$

2.3. Conductivity of Extrinsic Semiconductors when Intrinsic Charge Carrier Densities are neglected

- In the case of extrinsic semiconductors, the charge carriers available intrinsically is negligible when compared to the added impurity atoms
- This will modify the equations for conductivity of extrinsic semiconductors.

2.3.1. Conductivity of P Type Semiconductors

- Consider the equation (30) showing the conductivity of P type semiconductors.
- In this equation the hole density p_p has two components

- (a) Intrinsic hole density due to holes available in pure semiconductor
- (b) Hole density N_a contributed by the added acceptor impurity
- If we neglect the intrinsic hole density, the equation (30) will be modified as

$$\sigma = e N_a \mu_h \quad (35)$$

2.3.2. Conductivity of P-Type Semiconductors

- Consider the equation (34) showing the conductivity of N-type semiconductors.
- In this equation the hole density μ_h has two components
 - (a) Intrinsic electron density due to electrons available in pure semiconductor
 - (b) Hole density N_d contributed by the added donor impurity
- If we neglect the intrinsic electron density, the equation (34) will be modified as

$$\sigma = e N_d \mu_e \quad (36)$$

